

BraTS MRI Reconstruction Project

Generated from config `config/formal_train.yaml` on 2026-04-28 15:11:13.

Group Information

- Course: BME AI Project 1
- Group: 第 4 组
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- Note: This report was generated from the final verified experiment outputs and then reviewed before submission.

AI Usage Declaration

AI tools were used only as development assistance for code debugging, experiment orchestration, and draft editing. All implementation decisions, experimental validation, result interpretation, and final submitted materials were manually reviewed and confirmed by the student before submission.

Presentation reminder: AI assistance is explicitly declared in the generated presentation slides, in accordance with the assignment requirement.

Project Objective

This project reconstructs fully sampled T2 brain MRI slices from $AF=5$ undersampled k-space using the BraTS dataset. The work is organized into three assignment tasks: undersampling simulation, a baseline deep reconstruction model, and a multi-modal unrolled reconstruction model with data consistency.

Dataset and Preprocessing

- Dataset path: `../bmeaidataset`
- Modalities used: T1, T2
- Slice axis: 2

- Intensity normalization: z-score on non-zero voxels
- Split strategy: patient-level train/validation/test
- Slice selection for this run: 16 central slices per patient
- Volume loading mode: preloaded in RAM to reduce I/O stalls
- Split counts: train=6992, validation=1504, test=1504
- Data split unit: patient-level split to avoid leakage across adjacent slices from the same subject

Methods

Task 1

A 2D random variable-density sampling mask with acceleration factor 5 is generated in k-space. Fully sampled T2 slices are transformed with FFT, masked, and reconstructed with inverse FFT to obtain aliased images. This part establishes the artifact pattern that the learning-based models must remove.

Task 2

Task 2 uses a U-Net baseline with base channels 24, depth 4, batch size 8, MSE loss, and learning rate 0.0005. `ReduceLROnPlateau` is used for learning rate decay.

To improve runtime efficiency on the available RTX 4060 laptop GPU, the training pipeline uses slice caching, pinned memory, non-blocking GPU transfers, `channels_last`, TF32, and prefetch-friendly dataloading to reduce GPU idle time.

Task 3

Task 3 uses an unrolled U-Net with 2 cascades, data consistency layers, and multi-modal input consisting of aliased T2 plus fully sampled T1. The loss is **hybrid** with L1 weight 0.7.

The design motivation is that T1 provides stable anatomical structure, while the data-consistency layer constrains the network output to remain faithful to the measured undersampled k-space.

Division of Labor

- 唐志昊：完成项目整合、训练实验执行、结果核查、报告整理与提交。

Quantitative Results

- Task 2 improved PSNR from 25.76 dB to 35.18 dB and SSIM from 0.3628 to 0.9271.
- Task 3 improved PSNR from 25.76 dB to 38.12 dB and SSIM from 0.3628 to 0.9660.
- Compared with Task 2, Task 3 changed PSNR by 2.93 dB and SSIM by 0.0389.

Figures and Tables

Dataset Split Chart

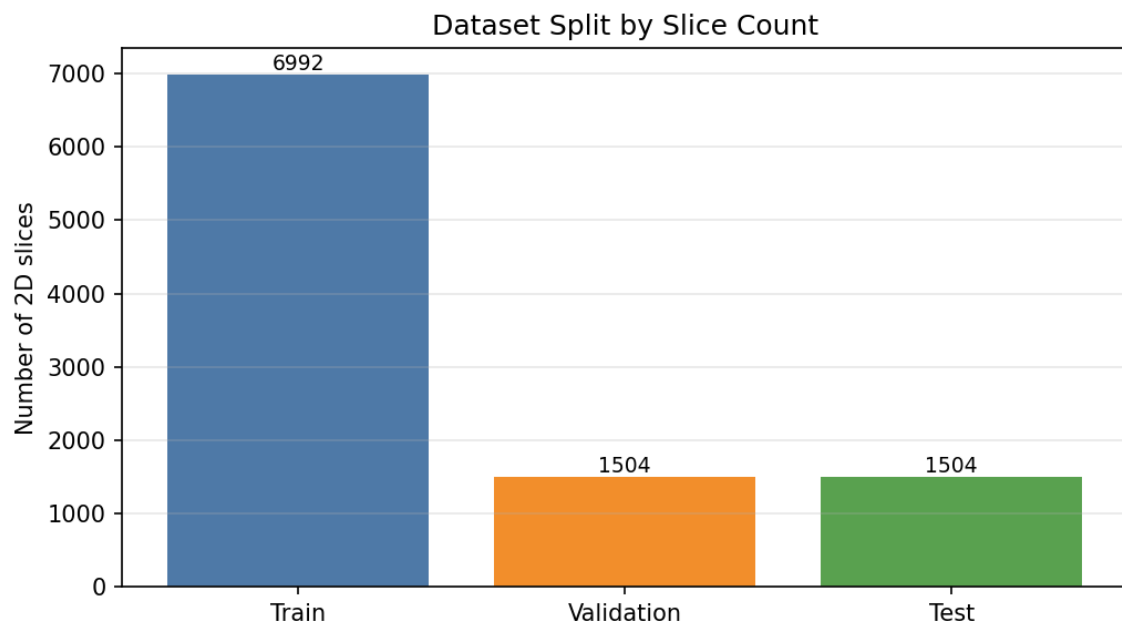


Figure 1: Dataset Split Chart

Metric Comparison Chart

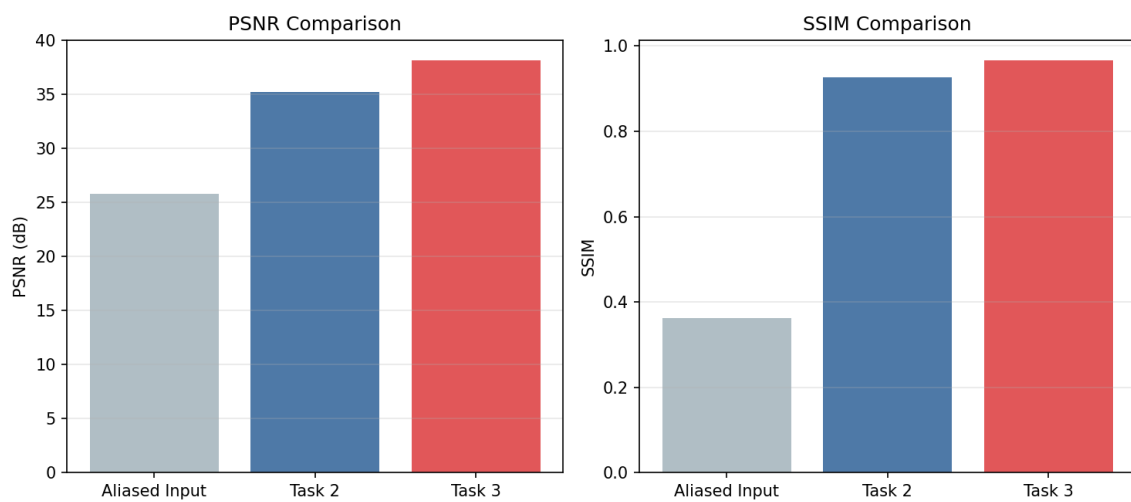


Figure 2: Metric Comparison Chart

Summary Tables

Dataset Split

Split	Slice Count
Train	6992
Validation	1504
Test	1504

Hyperparameters

Setting	Task 2	Task 3
Input	Aliased T2	Aliased T2 + full T1
Backbone	U-Net	Unrolled U-Net + DC
Base channels	24	16
Depth	4	3
Batch size	8	4
Epochs	10	8
Learning rate	0.0005	0.0001
Loss	MSE	hybrid

Quantitative Results

Method	PSNR (dB)	SSIM
Aliased input	25.76	0.3628
Task 2 baseline	35.18	0.9271
Task 3 multi-modal	38.12	0.9660

Improvements Over Aliased Input

Method	PSNR Gain (dB)	SSIM Gain
Task 2 baseline	9.43	0.5643
Task 3 multi-modal	12.36	0.6032

Task 3 Over Task 2

Comparison	Value
PSNR gain	2.93 dB
SSIM gain	0.0389

Worst-case Table

Rank	Patient	Slice	PSNR Before	PSNR After	SSIM Before	SSIM After
1	BraTS-GLI-00077-000	76	23.26	34.69	0.2865	0.9443

Rank	Patient	Slice	PSNR Before	PSNR After	SSIM Before	SSIM After
2	BraTS-GLI-00077-000	82	23.14	34.74	0.2875	0.9502
3	BraTS-GLI-00088-001	82	24.33	34.82	0.3160	0.9439
4	BraTS-GLI-00077-000	78	23.23	34.82	0.2942	0.9481
5	BraTS-GLI-00077-000	79	23.22	34.84	0.2921	0.9484

Task 1 Visualization

Task 2 Loss Curve

Task 2 Reconstructions

Task 3 Loss Curve

Task 3 Best Reconstructions

Task 3 Error Analysis

Error Analysis

The 5 worst-performing Task 3 cases are summarized below. Even in these difficult slices, the reconstructed outputs still remain substantially better than the aliased inputs, which indicates that the failure mode is degradation in relative quality rather than complete reconstruction collapse.

Rank	Patient	Slice	PSNR After	SSIM After
1	BraTS-GLI-00077-000	76	34.69	0.9443
2	BraTS-GLI-00077-000	82	34.74	0.9502
3	BraTS-GLI-00088-001	82	34.82	0.9439
4	BraTS-GLI-00077-000	78	34.82	0.9481
5	BraTS-GLI-00077-000	79	34.84	0.9484

Potential improvements for these cases include stronger edge-preserving loss terms, more cascades if runtime allows, and targeted inspection of slices with complex tumor boundaries.

Execution Status

- Current run status: Success
- Output directory: `outputs_formal`
- Total runtime: 24m 42s
- Last completed stage: assets
- Log file: `outputs_formal/logs/formal_pipeline.log`

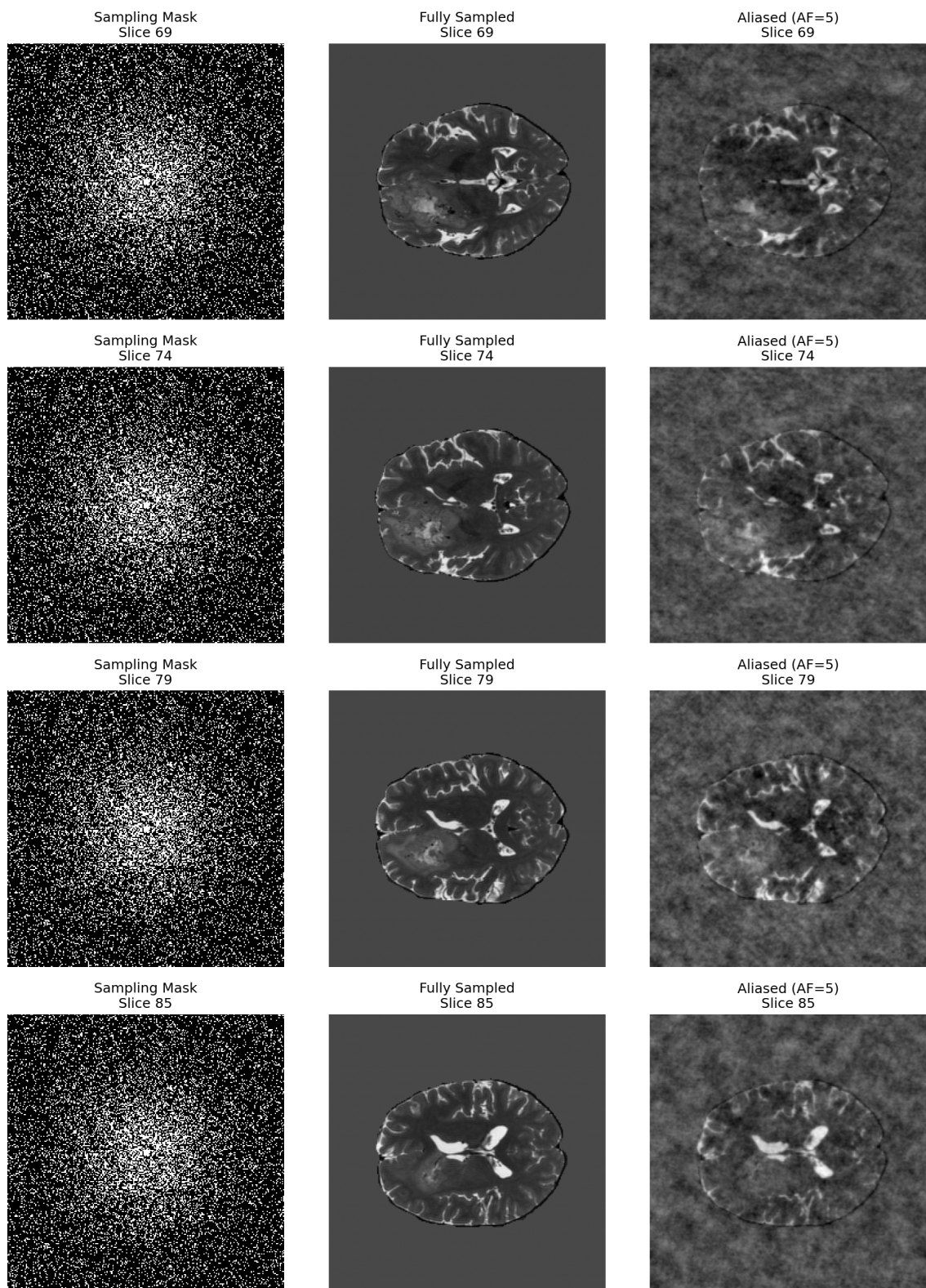


Figure 3: Task 1 Visualization

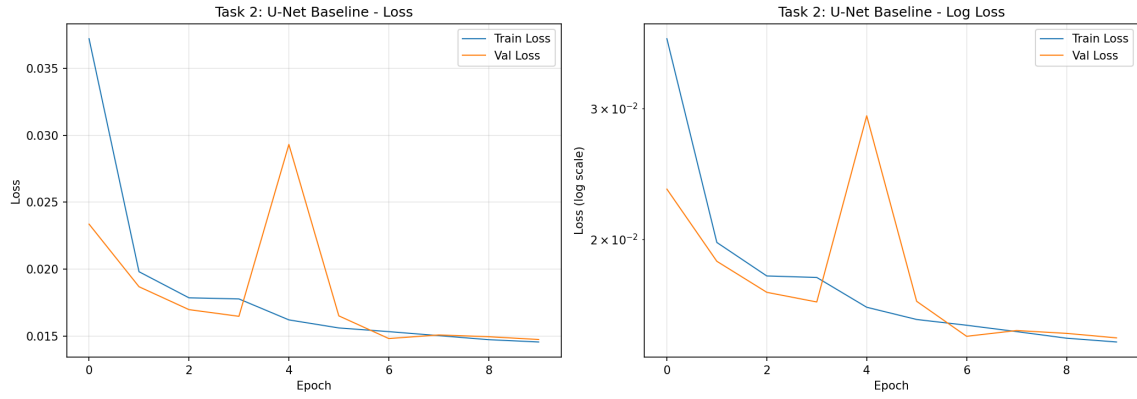


Figure 4: Task 2 Loss Curve

Discussion

The final experiment used 16 central slices per patient across all available BraTS patients and achieved a stable improvement over the aliased baseline in both tasks.

The Task 2 baseline recovered most of the missing image fidelity, improving PSNR by 9.43 dB and SSIM by 0.5643, which confirms that the basic supervised reconstruction pipeline converged correctly.

Task 3 further improved PSNR by 12.36 dB and SSIM by 0.6032 over the aliased input, and outperformed Task 2 by 2.93 dB PSNR and 0.0389 SSIM.

These results support the intended project conclusion: adding fully sampled T1 structural guidance together with unrolled data-consistency reconstruction yields clearer edges, fewer residual artifacts, and better quantitative fidelity than a single-modality baseline.

The remaining difficult cases are concentrated in slices with more complex local structure, suggesting that future improvements could come from stronger edge-aware losses, a deeper unrolled design, or targeted sampling and training strategies for harder anatomical regions.

Submission Checklist

- Code: prepared
- Report draft: `REPORT.md`
- LaTeX report: `REPORT.tex`
- PDF report: `REPORT.pdf`
- Presentation slides: generated as `slides.tex` and `slides.pdf`

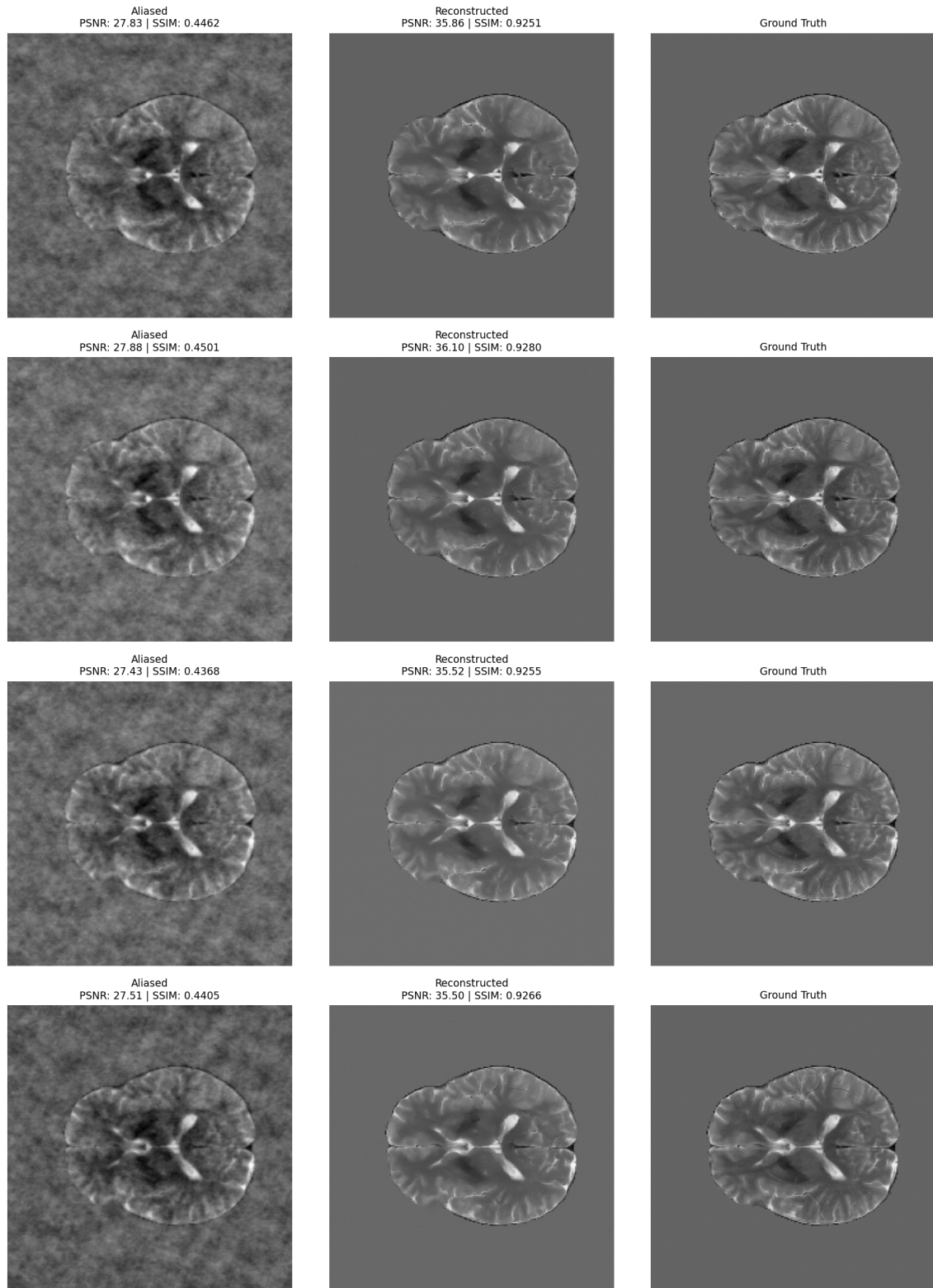


Figure 5: Task 2 Reconstructions

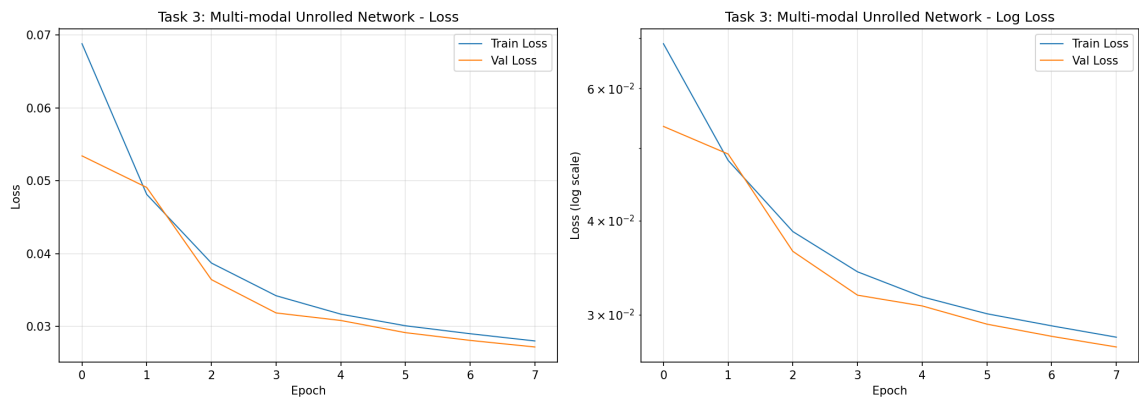


Figure 6: Task 3 Loss Curve

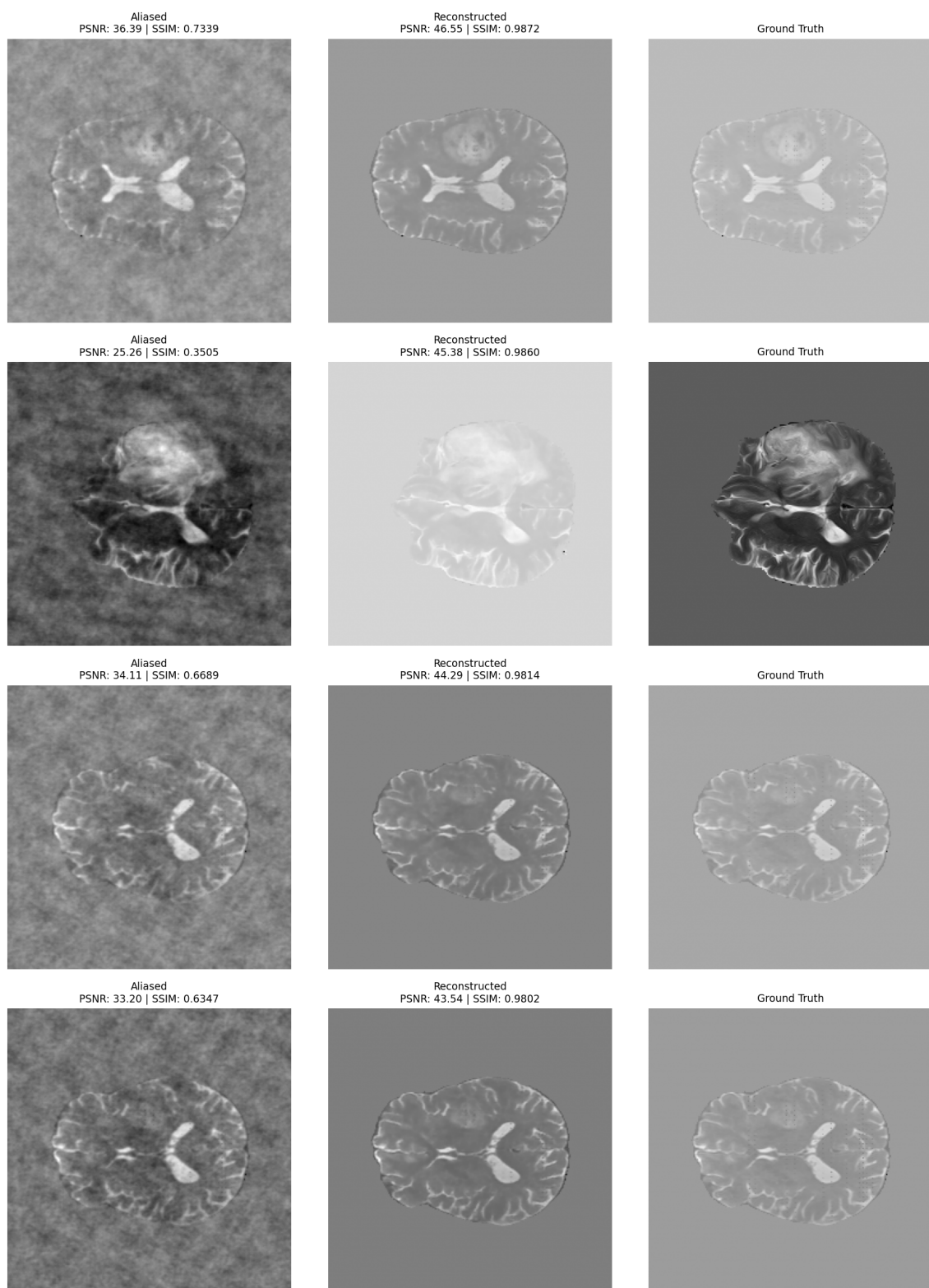


Figure 7: Task 3 Best Reconstructions

5 Worst Reconstruction Cases

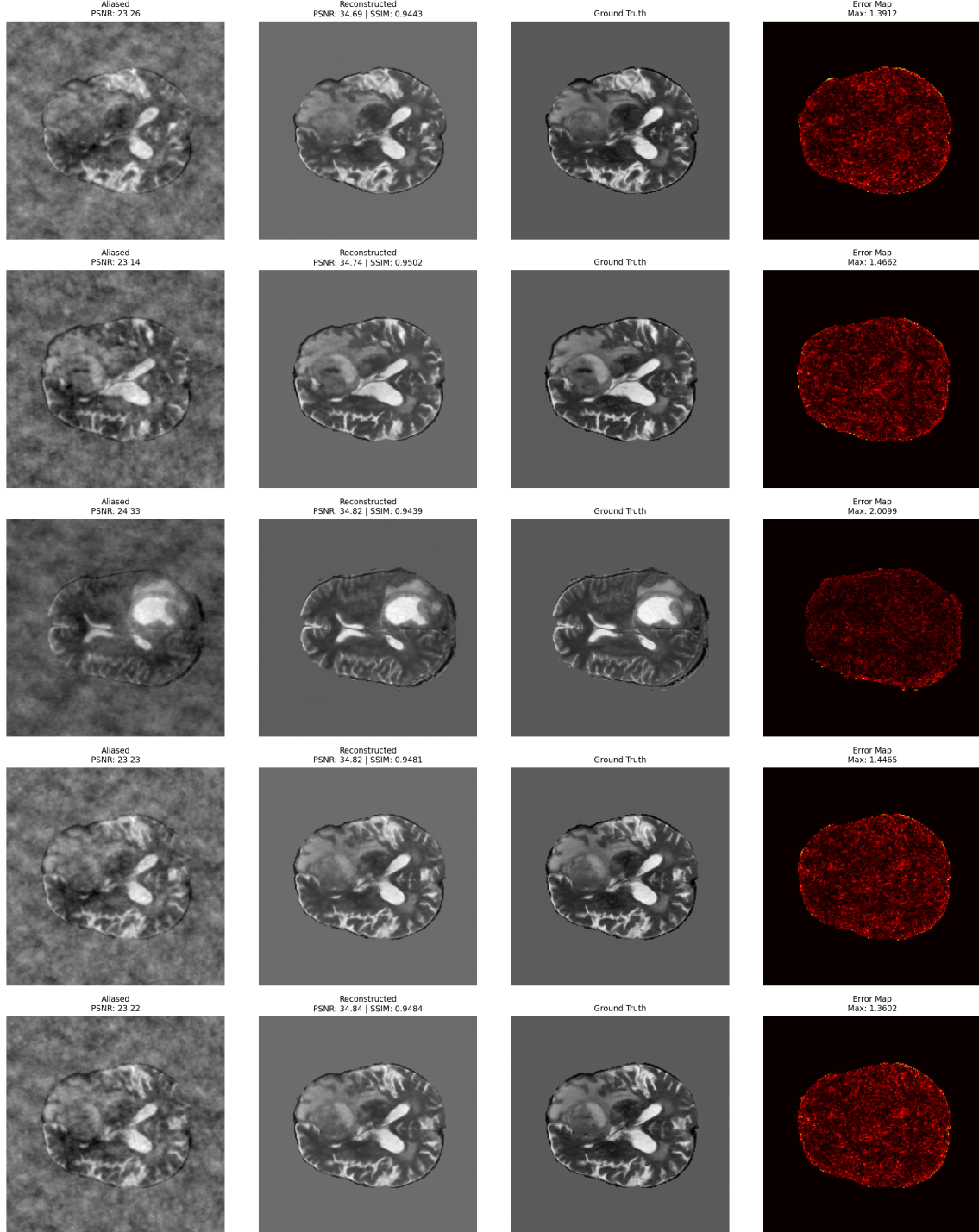


Figure 8: Task 3 Error Analysis